



BSC/S4565 Controller

User Manual

Honeywell S4565 Fireplace Control System
Version 2.0.1

GRS²

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Table of Contents

1. Introduction
2. Getting Started
3. Web Interface Overview
4. Home Page - Main Controls
5. Settings Page
6. Statistics Page
7. Event Log
8. Remote Control Setup
9. Home Automation Integration
10. Safety Features
11. Maintenance
12. Troubleshooting
13. Technical Specifications

1. Introduction

The BSC/S4565 Controller is an advanced smart control system designed for Honeywell S4565-based gas fireplaces. This ESP-based controller provides comprehensive management of your fireplace through multiple interfaces:

- Web-based interface accessible from any device
- IR/RF remote control support
- Home automation integration via Modbus TCP/IP HomeAssistant compatible
- Analog voltage control (0-10V)

1.1 Key Features

- Real-time monitoring of all safety systems
- Adjustable flame intensity with PWM valve control
- Automatic fan speed management with PDS support
- Wave function for natural flame effect
- Second valve/output control for dual-burner systems
- Comprehensive statistics and event logging
- Email alerts for faults and maintenance
- Over-the-air (OTA) firmware updates

⚠ WARNING: This controller manages gas-burning equipment. Always ensure proper installation by a qualified technician. Never bypass safety features during normal operation.

2. Getting Started

2.1 Initial Connection

When powered on for the first time, the controller creates its own WiFi network:

Setting	Value
Network Name (SSID)	BSC/S4565 Control - GRS²
Password	12345678
IP Address	192.168.4.1
Default Login Password	admin

2.2 Connecting to Your Home WiFi

1. Connect to the controller's WiFi network
2. Open a web browser and navigate to 192.168.4.1
3. Log in with password: admin
4. Go to the Settings page
5. Click 'Scan Networks' to find available WiFi networks
6. Select your network and enter the password
7. Click 'Connect' - the device will restart

i After connecting to your home WiFi, you can access the controller using its new IP address or via mDNS: <http://fireplace.local> (if supported by your network).

3. Web Interface Overview

The web interface provides full control and monitoring through five main pages:

Page	Description
Home	Main control panel - start/stop, fan control, valve adjustment
Settings	WiFi configuration, email alerts, maintenance, firmware updates
Statistics	Burning hours, start counts, charts, and usage history
Event Log	History of system events, faults, and actions
Remote	IR/RF remote control learning and configuration

4. Home Page - Main Controls

4.1 System Status

The status section displays real-time information about all safety systems:

Indicator	Meaning
Thermal Recoil	Overheat protection sensor status (Inactive = OK)
Limit Switch	Door/airflow safety switch (Closed = OK)
Fault Input	External fault signal from S4565 (Inactive = OK)
Start SSR	Main burner solid-state relay (ON = Burning)
PDS Status	Pressure Differential Sensor (Active = Fan auto-control)

4.2 Start/Stop Controls

Start Button: Initiates the fireplace startup sequence. The system performs safety checks before ignition.

Stop Button: Safely shuts down the fireplace. The exhaust fan continues running for a cool-down period.

Reset Button: Clears fault conditions after the underlying issue has been resolved.

4.3 Fan Controls

Exhaust Fan

Controls the speed of the exhaust/chimney fan (0-100%). When PDS is enabled, this is automatically managed to maintain proper pressure differential.

Output 2 / Second Valve

The second output can be configured in three modes:

- Mode 0 - Inlet Fan: PWM-controlled inlet fan speed
- Mode 1 - User Output: General purpose PWM output
- Mode 2 - Second Valve: On/off control for dual-burner systems

4.4 PWM Valve Control

The PWM valve controls flame intensity from 0-100%. This slider is only active when the fireplace is running.

Wave Function

When enabled, the flame intensity varies randomly to create a more natural, realistic fire appearance.

5. Settings Page

5.1 WiFi Configuration

Configure network connectivity:

- Scan Networks: Discover available WiFi networks
- Static IP: Optionally configure a fixed IP address
- AP Mode: Switch back to Access Point mode

5.2 Email Alerts

Configure automatic email notifications for:

- Fault detection
- Maintenance reminders
- Auto-shutdown events

Up to 3 email recipients can be configured. SMTP server settings pre-configured via our server. Notifications will be send from fireplaceATgrss.be

5.3 Maintenance Settings

Maintenance Interval: Set the number of operating hours between maintenance reminders (default: 300 hours).

Reset Maintenance: Clear the maintenance counter after service has been performed (requires admin password).

5.4 Auto-Shutdown

Configure automatic shutdown after a set number of hours of continuous operation. This is a safety feature to prevent unattended extended use.

5.5 Firmware Updates

Manual Update

Upload a firmware .bin file directly through the web interface.

OTA (Over-The-Air) Update

When connected to the internet, the system automatically checks for updates. If available, click 'Update Now' to download and install the latest firmware.

 **WARNING:** Do not power off the device during a firmware update. This may corrupt the firmware and require manual recovery

6. Statistics Page

The Statistics page provides comprehensive usage data and historical tracking.

6.1 Counters

Counter	Description
Burning Hours	Total hours of main burner operation
Pilot Burning Hours	Hours of pilot flame operation
Second Valve Hours	Total hours second valve/burner active
Start Count	Number of successful ignition cycles
Fault Count	Total number of fault events recorded

6.2 Charts

Interactive charts display the last 30 days of:

- Daily burning hours
- Start counts per day
- Average valve position
- Second valve usage (if applicable)

7. Event Log

The Event Log records all significant system events with timestamps. Events are stored in non-volatile memory and persist through power cycles.

7.1 Event Types

Event	Description
SYSTEM_START	Controller powered on/restarted
FIRE_ON / FIRE_OFF	Fireplace started or stopped
FAULT_DETECTED	Safety fault triggered
FAULT_RESET	Fault condition cleared
SECOND_VALVE_ON/OFF	Second valve state changed
WIFI_CONNECTED	Network connection established
FIRMWARE_UPDATE	Firmware update completed

8. Remote Control Setup

The controller supports both IR (infrared) and RF (radio frequency) remote controls.

8.1 Supported Remote Types

- IR Remote: Standard infrared remotes
- (RF Remote: Radio frequency remotes) under development
- AM05 Remote: Pre-configured

8.2 Learning Remote Codes

8. Navigate to the Remote page
9. Select your remote type (IR or RF)
10. Click 'Learn' next to the function you want to program
11. Press the desired button on your remote within 5 seconds
12. The system will confirm successful learning or show an error

8.3 Available Functions

- Start/Stop: Toggle fireplace on/off
- Flame Up: Increase flame intensity by 10%
- Flame Down: Decrease flame intensity by 10%
- Wave Activate: Toggle wave function
- Reset: Reset function in case of error

9. Home Automation Integration

9.1 Modbus TCP/IP

The controller provides a Modbus TCP server for integration with home automation systems. ModBus-Registers operations and details visible on UI

Setting	Value
Port	502 (standard Modbus TCP)
Slave ID	2

9.2 Analog Input Control

When enabled, the controller responds only to 0-10V analog signals:

Voltage Range	Action
0V - 1.8V	Stop fireplace
2.0V - 8.0V	Start and adjust flame intensity proportionally
> 8.0V	Enable wave function

10. Safety Features

The BSC/S4565 Controller incorporates multiple layers of safety protection.

10.1 Safety Systems

Thermal Recoil Protection

Monitors the fireplace temperature. If overheating is detected, the system immediately shuts down.

Limit Switch Monitoring

Ensures that access doors are closed and airflow paths are clear before allowing operation.

Fault Input Detection

Monitors the S4565 control module's fault output. Any fault condition triggers immediate shutdown.

PDS (Pressure Differential Sensor)

When enabled, automatically adjusts exhaust fan speed to maintain safe chimney draft.

10.2 LED Fault Indicators

The status LED indicates fault conditions through blink patterns:

Blink Pattern	Meaning
1 Blink	Thermal recoil fault
2 Blinks	Limit switch open
3 Blinks	Fault input active
4 Blinks	PDS failure
Solid On	System running normally
Breathing	Start Sequence

10.3 Safety Override

For diagnostic purposes, safety checks can be temporarily bypassed:

- Requires admin password
- Automatically disables after 5 minutes
- Should only be used by qualified technicians

 **WARNING:** Safety override bypasses critical protection systems. Only use during installation and troubleshooting by qualified personnel. Never leave unattended with safety override enabled.

11. Maintenance

11.1 Regular Maintenance Schedule

The controller tracks operating hours and reminds you when maintenance is due.
Recommended service interval: 300 hours (adjustable in Settings).

11.2 Maintenance Checklist

- Inspect and clean the exhaust fan
- Check all gas connections for leaks
- Clean the burner assembly
- Verify safety sensor operation
- Test all control functions
- Reset the maintenance counter after service

11.3 Resetting Maintenance Counter

13. Navigate to Settings page
14. Scroll to Maintenance section
15. Click 'Reset Maintenance Counter'
16. Enter admin password when prompted

12. Troubleshooting

12.1 Common Issues

Cannot Connect to Web Interface

- Verify you're connected to the correct WiFi network
- Try the direct IP address instead of mDNS hostname
- Power cycle the controller
- Check if AP mode is active (connect to BSC/S4565 Control network)

Fireplace Won't Start

- Check status indicators for fault conditions
- Verify all safety sensors show OK status
- Try the Reset button if a fault is indicated
- Check the Event Log for error details

Remote Control Not Working

- Verify remote type matches settings (IR/RF)
- Re-learn the remote codes
- Check remote battery
- Ensure clear line-of-sight for IR remotes

Flame Intensity Not Adjusting

- Fireplace must be running to adjust flame
- Verify PWM valve wiring

12.2 Factory Reset

If all else fails, perform a factory reset:

17. Navigate to Settings page
18. Scroll to bottom and click 'Full Reset'
19. Confirm the reset
20. Device will restart with default settings

⚠ WARNING: Factory reset erases ALL settings including WiFi configuration, learned remote codes, and statistics. Use only as a last resort.

13. Technical Specifications

Parameter	Specification
Processor	ESP Dual-Core
WiFi	802.11 b/g/n 2.4GHz
PWM Outputs	3x 8-bit (Exhaust, Inlet/Valve, PWM Valve)
Digital Inputs	Thermal, Limit Switch, Fault, PDS
Analog Input	0-10V (via voltage divider)
IR Receiver	38kHz compatible
RF Receiver	(433MHz)
Communication	Modbus TCP/IP (Port 502)
Storage	EEPROM + LittleFS
Statistics Retention	30 days daily history
Watchdog Timer	60 second timeout
Laagspanningsrichtlijn 2014/35/EU (Low Voltage Directive) - **EMC-richtlijn 2014/30/EU** (Electromagnetic Compatibility) - **RoHS-richtlijn 2011/65/EU** (Restriction of Hazardous Substances) - **Radio Equipment Directive 2014/53/EU** (voor WiFi-functionaliteit) EN 60204-1:2018 - Veiligheid van machines - Elektrische uitrusting - EN 61010-1:2010 - Veiligheid elektrische meet-, regel- en laboratoriumapparatuur EN 55032:2015 - Multimedia apparatuur - EMC-eisen - Emissie - EN 55035:2017 - Multimedia apparatuur - EMC-eisen - Immuniteit - EN 301 489-1 V2.2.3 - EMC voor radio-apparatuur - Algemene eisen - EN 301 489-17 V3.2.4 - EMC voor breedband datatransmissiesystemen EN 300 328 V2.2.2 - 2,4 GHz breedband transmissiesystemen - EN 62479:2010 - SAR-beoordeling draadloze communicatieapparatuur EN 50581:2012 - Documentatie voor RoHS-conformiteitsbeoordeling	

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